

Predicting EU Transitions across Pre-COVID, COVID, and Post-COVID Periods

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1. Introduction and Motivation

Determining which populations are more susceptible to job loss and how economic crises affect these populations is key to the design of unemployment insurance, workforce retraining programs, and social insurance policy. This paper aims to investigate how the structure of employed-to-unemployed (EU) transition risk changed across three distinct macroeconomic periods: a stable pre-pandemic economy, the COVID-19 shock, and the post-pandemic recovery, and which worker characteristics are most predictive of this transition risk in each period. Using individual-level monthly data from the Current Population Survey (CPS) linked via unique person identifiers, we are able to construct a panel of month-to-month employment transitions for over 2.2 million person-month observations and model the probability that a currently employed worker becomes unemployed in the following month.

We hypothesize that the global pandemic did not uniformly increase the level of job-loss risk, but instead selectively amplified risk for workers in certain demographic, educational, and occupational groups—particularly those in part-time arrangements, lower-education brackets, and industries most exposed to social-distancing constraints—while the post-pandemic recovery largely restored the pre-pandemic structure of risk.

We first estimate logistic models separately by period and compare coefficients and marginal effects to assess structural stability. We then evaluate out-of-sample predictive performance using ROC-AUC and precision-recall AUC. Finally, we benchmark a range of classification methods, including discriminant analyses, tree-based models, boosting, support vector machines, and k-nearest neighbors.

Our main findings indicate that job-loss risk rises sharply during COVID, with the monthly employed-to-unemployed transition rate nearly tripling before returning to pre- pandemic levels in the recovery period. Job-loss risk remains strongly heterogeneous across workers, with lower education, younger age, and part-time status consistently associated with higher transition probabilities. Importantly, the structure of this risk shifts during the pandemic, as the predictive relationships between worker characteristics and job loss change across periods, reflecting the uneven impact of the shock. While our models capture meaningful variation in risk, predictive performance is constrained by the rarity of unemployment transitions. These findings are robust to alternative definitions of job loss, including transitions out of the labor force.

2. Literature Review

The foundational framework for understanding job loss as a market flow comes from Shimer (2012), who divides unemployment dynamics into a job-finding rate and a job-separation rate. He finds that separation rates are largely acyclical and that most unemployment changes are caused by changes in the job-finding rate, rather than large spikes in layoffs. This serves as an important benchmark for our analysis as if Shimer's results were to hold universally, the structure of job-loss risk across worker characteristics would be roughly stable over time, and a model estimated in one period would generalize well to other periods. However, our results challenge this in the specific context of large aggregate shocks as during the COVID-19 period, the EU transition rate nearly tripled, a magnitude that is inconsistent with the acyclical separation rate and consistent with the view that extreme events can cause a different layoff regime.

A second key strand of literature that our paper builds upon refers to the heterogeneity in job-loss risk across workers. Jacobson, LaLonde, and Sullivan (1993) provide the foundational evidence that displacement is not random across workers, but rather concentrated among those with less firm-specific capital and in more exposed industries and that its consequences are persistent. Autor (2015) extends this by documenting that lower-skilled workers or those with less education are more exposed to disruptions in the labor market, such as what we experienced during COVID-19. Haltiwanger et al. further claim that “younger workers more likely flow into nonemployment during downturns”.³ Our results confirm the above findings, with the job-loss probability being amplified during COVID.

Furthermore, the COVID-19 pandemic itself produced large amounts of empirical literature documenting its distinctive shock to the labor market. Dingel and Neiman (2020) showed that the capacity to work remotely is sharply stratified by education and occupation, with college-educated and professional workers far better positioned to maintain employment through remote deployment. Additionally, Alon et al. (2020) document the effect of the pandemic on working women, particularly those in service and caregiving occupations, who faced disproportionate job losses during COVID, partly due to school and childcare closures disrupting labor supply. Both patterns are visible in our data and inform our interpretation of the shifting average marginal effects across periods.

This paper contributes to existing literature by exploring how the relationship between worker characteristics and job-loss risk change across the three different periods of the pandemic. We will see if these features remain stable during these large economic disruptions. While prior work has largely focused on aggregate dynamics or average effects over long time horizons, we take a micro-level, period-specific approach. In doing so, our contribution is twofold. First, we

provide evidence on how the separation rate remains approximately acyclical in normal periods but becomes strongly cyclical during extreme events such as the COVID-19 pandemic. Second, we document how the predictive importance of key worker characteristics changes across economic regimes, allowing us to move beyond average effects and instead characterize how the distribution of job-loss risk evolves in response to large-scale shocks.

3. Model

3.1 Behavioral and Informational Assumptions

We model job loss as the outcome of interactions between worker characteristics and labor demand. We treat CPS-observed attributes—education, age, sex, marital status, part-time status, and class of worker—as predetermined over the one-month horizon, consistent with standard worker displacement frameworks. This assumption is standard in the literature on worker displacement and is reasonable given the short horizon of our outcome variable.

We also assume employers' layoff decisions depend on worker characteristics and vary with macroeconomic conditions. In normal times, layoffs are driven by idiosyncratic firm shocks and are weakly correlated across workers, while in large downturns like COVID-19, sector-wide demand collapses generate correlated layoffs within industries. As a result, the direction of effects may remain stable, but their magnitudes shift as the structure of economic risk changes.

3.2 Period Definitions and Identification

We define the three periods—Pre-COVID (January 2018–February 2020), COVID (March–December 2020), and Post-COVID (January 2021–December 2023)—by the aggregate

labor market shock as opposed to traditional calendar dates due to the sharp discontinuity in March 2020, which provides a natural break point. Defining periods this way means that our Pre-COVID and Post-COVID estimates capture the baseline structure of risk under relatively normal labor market conditions, while the COVID period captures the shock regime.

We do not pursue a difference-in-differences design because we lack a clean control group: the COVID shock affected virtually all industries and regions simultaneously. Instead, our identification is based on period-by-period structural comparisons. We are not making causal claims about the effect of COVID on individual job-loss risk, but rather characterizing how the conditional distribution of job-loss risk changed across regimes and which observable characteristics drove that change.

4. Data Description

4.1 Data Source and Sample Construction

Our data comes from the Integrated Public Use Microdata Series (IPUMS) Current Population Survey (CPS), which provides harmonized individual-level monthly survey microdata. The CPS is a monthly household survey of approximately 60,000 interviewed households, conducted by the U.S. Census Bureau for the Bureau of Labor Statistics, and is the official source of U.S. unemployment statistics. We use the Basic Monthly CPS files for January 2018 through December 2023, yielding an initial extract of 8,856,323 person-month records.

The CPS employs a rotating panel design in which sampled households are interviewed for four consecutive months, rotated out for eight months, and then interviewed for four additional months. Crucially, IPUMS provides the variable CPSIDP, which uniquely identifies a person

across their rotation appearances and allows us to link individual records across consecutive months. We exploit this design to construct month-to-month employment transition records: for each person observed as employed in month t , we link to their employment status in month $t+1$, verifying that the two records are exactly one calendar month apart.

We restrict the sample to working-age adults between 18 and 64 years old and drop records with $EMPSTAT = 0$ (not in universe, typically non-civilian or non-respondents) or $CPSIDP = 0$ (records that cannot be linked across months). After these cleaning steps, the full CPS sample contains 5,216,371 person-month observations. From these, we retain only those observations where the individual is employed in month t and has a valid consecutive link to month $t+1$, and where the month $t+1$ employment status is either employed or unemployed. We exclude transitions to Not in the Labor Force (NILF), as these reflect labor force withdrawal (e.g., retirement, caregiving, disability) rather than involuntary layoff, which is the economic phenomenon of interest. We conduct a robustness check in Section 6 verifying that this exclusion does not materially affect our findings.

The resulting analysis panel contains 2,211,374 person-month observations across the three periods: 928,079 observations in the Pre-COVID period, 275,613 in the COVID period, and 1,007,682 in the Post-COVID period.

4.2 Variable Construction

Our outcome variable is a binary indicator equal to 1 if the worker transitions from employment at month t to unemployment at month $t+1$ (EU transition = 1) and 0 if the worker remains employed (EU transition = 0). Employment status is coded from the CPS variable

EMPSTAT values 10 and 12 which correspond to employed, while values 20, 21, and 22 correspond to unemployed.

Our predictor variables capture key demographic and job characteristics. Female is a binary indicator equal to 1 for women ($SEX = 2$). Married is a binary indicator equal to 1 for individuals who are married with a spouse present ($MARST = 1$), capturing household insurance effects on job-loss risk. Part-time is a binary indicator equal to 1 if usual hours worked per week ($UHRSWORKT$) are less than 35; observations with $UHRSWORKT = 999$ (hours vary) are treated as missing for this variable.

Education is coded into five ordered categories from the CPS EDUC variable: less than high school ($EDUC < 60$), high school diploma ($60 \leq EDUC < 73$), some college ($73 \leq EDUC < 110$), bachelor's degree ($110 \leq EDUC < 123$), and graduate degree ($EDUC \geq 123$). In regression models, high school diploma serves as the reference category. Age is grouped into five intervals (18–24, 25–34, 35–44, 45–54, 55–64), with ages 35–44 as the reference group. Worker class is derived from the CLASSWKR variable and distinguishes private sector employees, government employees (federal, state, and local), non-profit workers, self-employed workers, and a residual other category; private sector workers serve as the reference in regression models.

4.3 Descriptive Statistics

Worker composition is largely stable across periods, reflecting the fact that we are sampling from a roughly consistent population of employed workers. The female share remains near 46%, the married share near 54%, and the part-time share near 11% across all three periods. The educational composition is also stable, with approximately 25% of employed workers holding a bachelor's degree and 14–16% holding a graduate degree. The slight increase in the share of

college-educated workers over the 2018–2023 period reflects secular trends in educational attainment in the workforce.

Figure 1 presents the overall EU transition rate as we can see the striking three-fold increase in EU transition rate from 0.92% pre-COVID to 2.65% during peak COVID periods, followed by a near-complete reversion to 0.93% post-COVID. This pattern confirms that the COVID period represents a discrete and temporary regime shift in aggregate job-loss risk as opposed to a prolonged structural change.

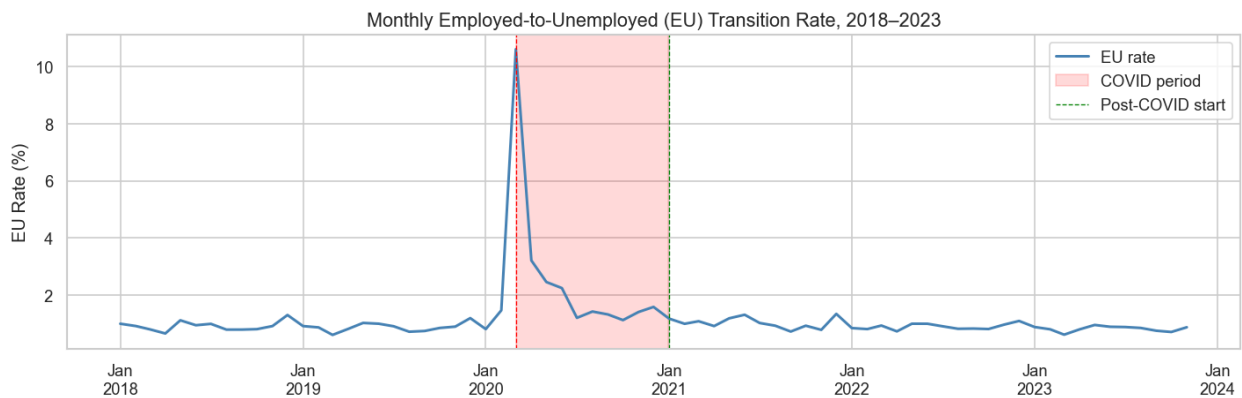


Figure 1: Monthly Employed-to-Unemployed (EU) Transition Rate, 2018-2023

Table 1 presents EU transition rates disaggregated by key subgroups across the three periods and gives us further insight into the mechanics of Figure 1. First, pre-COVID, the education gradient was already pronounced and persistent as graduate-degree holders had an EU rate of 0.47%, compared to workers with only a high school diploma who had an EU rate of 2.13%. During COVID, this gap widened substantially as the high school diploma group's EU rate rose to 5.04%, while the graduate degree group's rate only rose to 1.25%. Post-COVID, the gradient returned close to pre-pandemic levels. Second, part-time workers are consistently more vulnerable than full-time workers, with the EU rate being approximately ~2.3x larger across all three periods.

Third, gender differences are modest in the pre-COVID and post-COVID periods, but the female EU rate actually exceeded the male rate during COVID (2.87% vs. 2.44%), which can be explained due to the disproportionate concentration of women in contact-intensive service sector occupations. Fourth, government workers face substantially lower job-loss risk than private sector workers across all periods, with the government advantage actually shrinking during COVID as some public-sector layoffs occurred due to fiscal pressures.

Table 1: EU Transition Rate by Worker Characteristics and Period			
Subgroup	Pre-COVID	COVID	Post-COVID
Education			
Less than High School	1.85%	4.39%	2.02%
High School Diploma	2.13%	5.04%	2.08%
Some College	1.07%	3.27%	1.11%
Bachelor's Degree	0.60%	1.87%	0.62%
Graduate Degree	0.47%	1.25%	0.45%
Sex			
Male	0.99%	2.44%	0.99%
Female	0.82%	2.87%	0.86%
Part-Time Status			
Full-Time	0.78%	2.20%	0.78%
Part-Time	1.95%	4.90%	1.98%
Worker Class			
Private Sector	1.07%	3.28%	1.14%
Government	0.37%	1.34%	0.38%
Non-Profit	0.54%	1.64%	0.55%
Self-Employed	0.70%	2.48%	0.73%

Note: EU Transition rates computed as the share of person-month observations in each subgroup in which the employed worker transitions to unemployment in the following month. Some colleges are in the modal education category and account for approximately 54% of observations.

4.4 Data Limitations

While the CPS provides a large sample size and thorough data, there are still limitations that deserve acknowledgment. First, the CPS is a repeated cross-section with a limited panel dimension. This means that each household is observed for at most eight months, and the rotation design makes it so that approximately one-quarter of the employed sample in any given month cannot be linked to the following month. Unlinked observations are excluded from the analysis, which could introduce selection bias if linkage failure is correlated with job-loss risk.

Second, the CPS measures labor force status based on self-report in a reference week, which means that the EU transition indicator captures status changes that the respondent reports retrospectively. Workers who lost a job and quickly found a new one within the month would not appear as EU transitions in our data; our outcome more precisely measures persistent job-loss spells that extend across consecutive monthly interviews.

Third, we do not observe wages, earnings, or job tenure in this dataset (wage data are available only in the March Annual Social and Economic Supplement, which is not linkable in the same way). Job tenure and wage level are known predictors of layoff risk, and their omission means our estimates should be interpreted as reduced-form associations rather than structural parameters.

5. Econometric and Machine Learning Specifications

5.1 Primary Logistic Regression Specification

The primary empirical specification is a logistic regression of the EU transition indicator on a vector of worker characteristics, which are estimated separately for each of the three periods. For worker i observed in period p , our model is:

$$Pr(EU_{i,t+1} = 1 \mid X_{i,t,p}) = \Lambda(\beta_{0,p} + \beta_{1,p}female_i + \beta_{2,p}married_i + \beta_{3,p}part_time_i + \beta_{4,p}educ_i + \beta_{5,p}cls_i + \beta_{6,p}age_i)$$

where $\Lambda(\cdot)$ denotes the logistic CDF and p indicates that all parameters are estimated independently within each period. The vector $educ_i$ contains dummies for the four education categories with high school diploma as the reference group. The vector cls_i contains worker classes with the private sector as the reference. The vector age_i contains dummies for age groups with ages 35-44 as the reference group.

The key parameters of interest are the period specific coefficient vectors β_p , and the differences in β_p across periods reveal whether the predictive content of each characteristic changed with the macroeconomic periods. For example, a larger magnitude of the part-time coefficient during COVID relative to pre-COVID indicates that part-time status became a stronger predictor of layoff risk during the pandemic. To aid interpretation, we report average marginal effects (AMEs), which measure the average change in transition probability associated with each characteristic in percentage-point terms and are directly comparable across periods despite differing baseline rates.

5.2 Predictive Specification and Evaluation Metrics

For the predictive component of the analysis, we re-estimate the same logistic model using scikit-learn with L2 regularization ($C = 1.0$) and `class_weight = 'balanced'` to address severe class imbalance: EU transitions represent approximately 0.92–2.65% of observations depending on the period. The balanced weighting up-weights minority-class ($EU = 1$) observations during training by a factor proportional to the class ratio, which improves the model's ability to discriminate at the relevant margin. All models are evaluated on a held-out test set consisting of a stratified random 20% sample of each period, ensuring the EU rate of the test set matches that of the training set.

We report two primary predictive metrics. The area under the ROC curve (ROC-AUC) measures the probability that the model assigns a higher predicted probability to a randomly chosen $EU = 1$ observation than to a randomly chosen $EU = 0$ observation, with a value of 0.5 corresponding to random guessing and 1.0 to perfect discrimination. While ROC-AUC is the most widely reported classification metric, it can be misleading under severe class imbalance because it averages performance across all classification thresholds, including many that are irrelevant in practice. We therefore also report the area under the precision-recall curve (PR-AUC), which focuses on the model's ability to correctly identify the minority class. Under random prediction, PR-AUC equals the baseline EU rate (~ 0.01), providing a more stringent benchmark. We additionally report the Brier score (mean squared error of predicted probabilities), top 1% precision (the fraction of true EU transitions among the 1% of observations assigned the highest predicted probability), and top 5% precision, which are directly relevant to any policy application targeting high-risk workers.

For the time-based robustness check, we split each period chronologically rather than randomly, where the first portion of each period constitutes the training set, and the latter portion

the test set. Specifically, for Pre-COVID, the training set covers January 2018-December 2019, and the test set covers January-February 2020. For COVID, the training set covers March-June 2020, and the test set covers July-December 2020. For Post-COVID, the training set covers 2021-2022, and the test set covers 2023. The time-based split evaluates the model's ability to predict job-loss risk prospectively, which is the operationally relevant question for policy applications, and avoids the potential information leakage that could arise if future observations influence training.

5.3 Additional ML Classifiers

To assess whether more flexible methods improve upon the logistic regression baseline, we apply seven additional classifiers to the same EU transition prediction task: Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), Decision Tree, Random Forest, Gradient Boosting, and K-Nearest Neighbors (KNN), alongside a Support Vector Machine (SVM). All models are fit to the same feature matrix X and evaluated under the same 80/20 stratified train-test split, estimated separately within each period. This allows a direct comparison against the logistic regression baseline in Section 5.2, with any difference in performance attributable to the algorithm rather than to differences in data or evaluation design.

5.4 Assumptions

Our identification rests on the key assumption that, after accounting for the included variables (such as observable characteristics), unobserved worker traits are not correlated with the likelihood of transitioning into EU status within a given time period. Since the same workers are not tracked over time, the dataset is too short to support that approach, so the analysis instead

compares different individuals at a single point in time. As a result, it relies on a cross-sectional conditional independence assumption, meaning that comparisons across individuals are valid after controlling for the observed variables. A limitation of this approach is the absence of data on wages and job tenure, both of which influence layoff risk. By omitting these factors, bias may be introduced into the results. However, the consistency of findings before and after COVID suggests that such missing variables are unlikely to significantly distort comparisons across time periods.

6. Results

6.1 Logistic Regression

The logistic regression baseline achieves ROC-AUC of approximately 0.68 Pre-COVID, 0.73 during COVID, and 0.68 Post-COVID. These scores confirm that even the simplest model in our comparison captures meaningful predictive signals, particularly during the COVID period where the concentration of job loss among observable worker groups is strongest. The COVID improvement of roughly 0.06 ROC-AUC points above the pre-COVID baseline is itself an important finding as it confirms our hypothesis that the pandemic did not merely uniformly raise the level of job-loss risk, but made it structurally more predictable from demographic and occupational characteristics.

After adding PR-AUC, we found that while precision-recall performance is low in absolute terms, that is to be expected in highly imbalanced settings where the positive class is rare. Figure 2 reports calibration results, demonstrating the closeness of predicted probabilities with realized event frequencies across risk bins. These results indicate that the model captures meaningful

signals in worker characteristics but faces inherent limits imposed by rare events and the coarseness of available predictors.

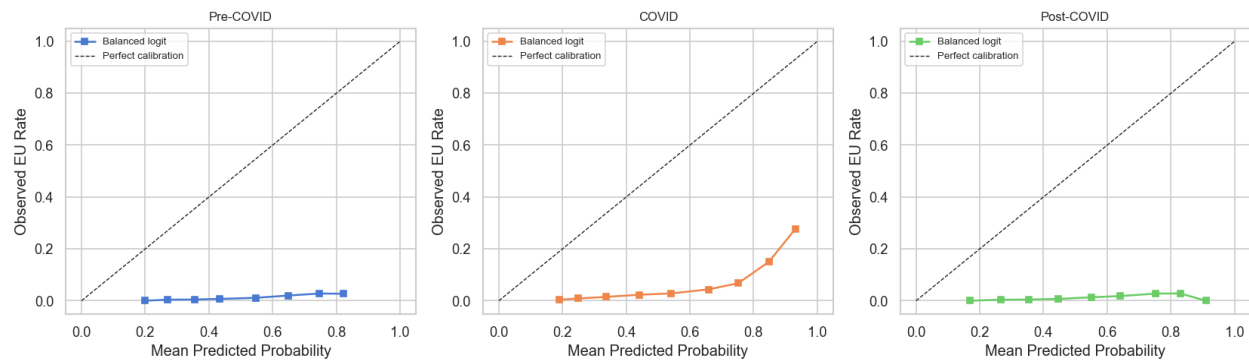


Figure 2: Calibration Curves by Period

(Note: The dashed 45-degree line indicates perfect calibration; deviations reflect probability miscalibration in this rare-event setting)

6.2 Average Marginal Effects and Coefficients

We find evidence that some relationships shift during COVID. The coefficient comparison in Figure 3 shows that the direction and magnitude of certain predictors differ notably across periods. For example, the estimated effect of being female differs across periods that female workers are associated with lower EU risk in pre- and post-COVID periods, but higher EU risk during COVID. This pattern is consistent with the situations of pandemic sectoral disruptions and differential caregiving constraints that may have affected women's employment stability disproportionately, reflective of Alon et. al's previous findings.

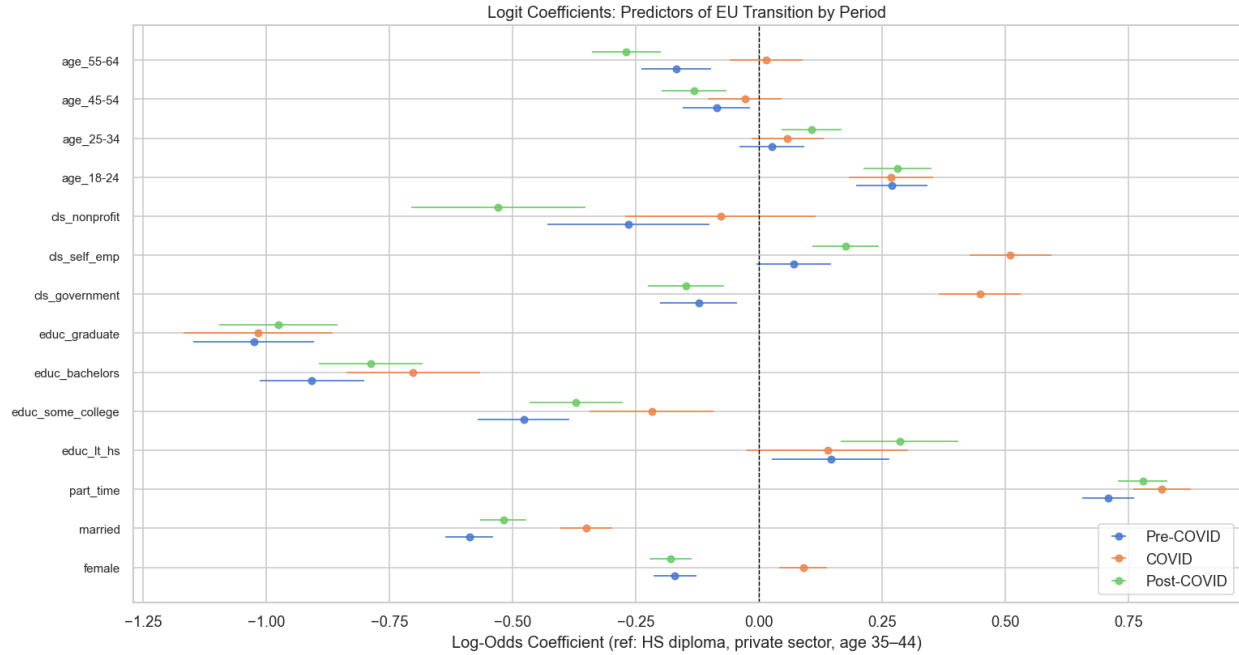


Figure 3: Logit Coefficients: Predictors of EU Transition by Period

We also observe obvious period-dependent patterns by class of worker. During COVID, some categories exhibit much larger increases in predicted EU probability than in pre- and post-COVID periods. This is especially visible in the AME heatmap in Figure 4, which summarizes the relative strength of each predictor across periods. Generally, the logit results support the hypothesis that the macroeconomic environment affects not only the average job-loss risk but also which worker characteristics are most predictive of near-term unemployment transitions. Again, we confirm the results of previous findings as we find that the average marginal effects for part-time, younger, and lower-educated individuals are higher.

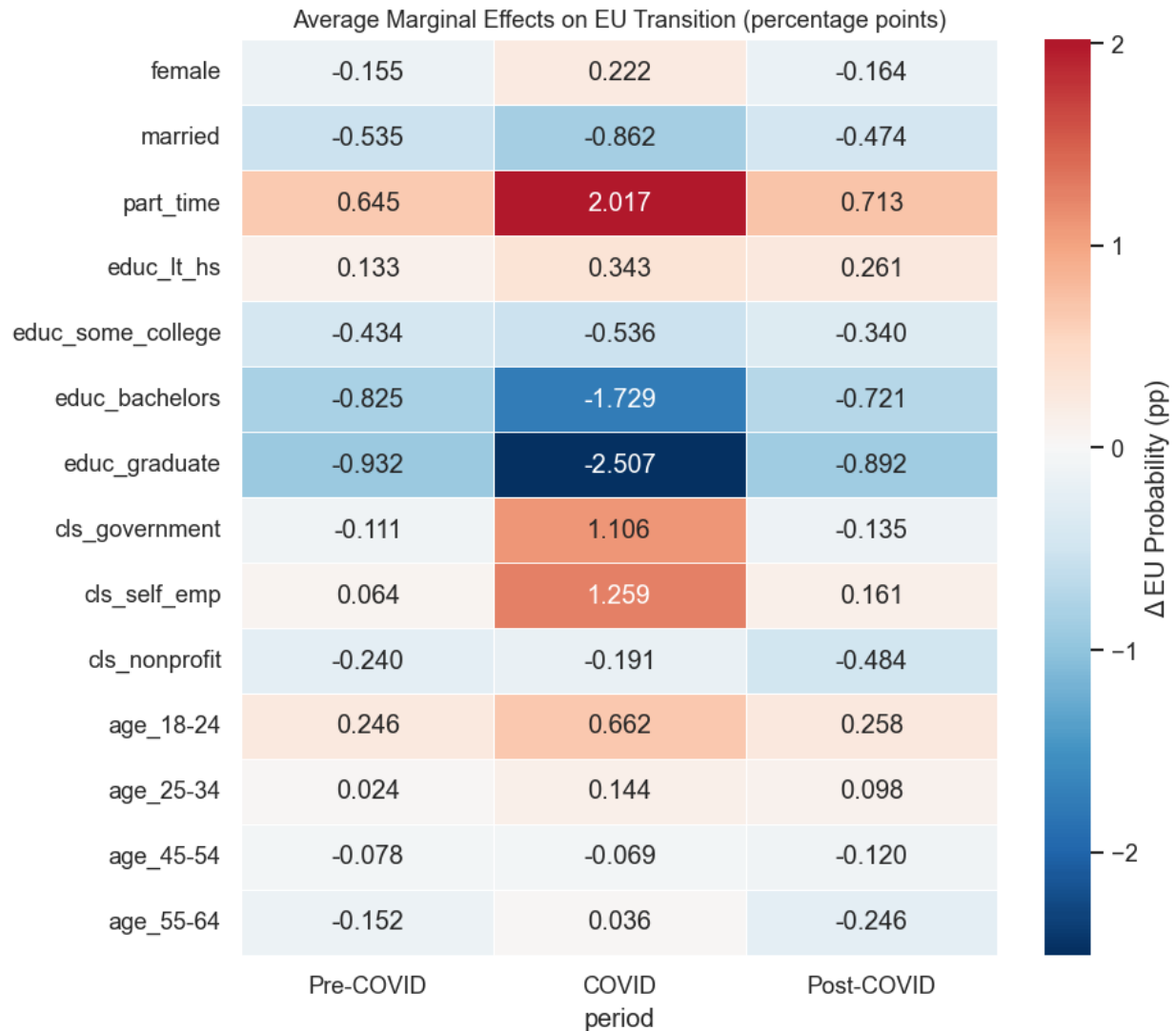


Figure 4: Average Marginal Effects on EU Transition

6.3 Additional Classifiers

The additional classifiers used to predict EU transition rates improve on logistic regression, but only marginally and for certain methods. Gradient Boosting is the only classifier that consistently outperforms logistic regression across all three periods, with ROC-AUC gains of approximately 0.01 points pre-COVID, 0.01 COVID, and 0.015 post-COVID. LDA performs

nearly identically to logistic regression in every period, which is expected given that both rely on linear decision boundaries. QDA and Decision Tree perform slightly below logistic regression in most periods. Random Forest underperforms logistic regression in all three periods, likely due to bagging suppressing the weak minority-class signal under severe class imbalance. SVM and KNN perform substantially worse, though their 30,000-observation training subsample is a confounding factor. What we can gather from these results is that logistic regression is a near-optimal linear classifier for this problem as the non-linear flexibility of ensemble methods offers only marginal additional predictive power, indicating that the relationship between worker characteristics and EU transition risk is largely linear and well-captured by the baseline specification in Section 5.1.

Across all eight classifiers, every model performs best during the COVID period. This is not a property of any particular algorithm but a feature of the data itself: the COVID-19 shock concentrated job loss among workers with specific, observable characteristics, part-time status, lower educational attainment, younger age groups, and exposure to contact-intensive industries, all of which are captured in the feature matrix. The classifiers can therefore assign elevated predicted risk to precisely the workers who were most vulnerable. In contrast, job loss in normal labor market conditions is driven largely by firm-level decisions and individual circumstances not captured in demographic survey data, producing weaker signals that no classifier can reliably detect. The Post-COVID period slightly outperforms Pre-COVID for most models, suggesting some lingering structural concentration of risk during the recovery phase. Table 2 summarizes the

out-of-sample performance of all eight classifiers by period under the 80/20 stratified random split, reporting ROC-AUC and PR-AUC along with test-set size.

Table 2: Classifier Performance by Period (80/20 Stratified Split)				
Period	Model	ROC-AUC	PR-AUC	N (Test)
COVID	Logistic Regression (baseline)	0.7347	0.1130	5,512
COVID	Gradient Boosting	0.7413	0.1122	5,512
COVID	LDA	0.7346	0.1130	5,512
COVID	QDA	0.7302	0.1033	5,512
COVID	Decision Tree	0.7278	0.1018	5,512
COVID	Random Forest	0.7149	0.1029	5,512
COVID	SVM (30k subsample)	0.6622	0.0504	5,512
COVID	KNN (30k subsample)	0.5965	0.0450	5,512
Post-COVID	Logistic Regression (baseline)	0.6838	0.0191	20,153
Post-COVID	Gradient Boosting	0.6926	0.0205	20,153
Post-COVID	LDA	0.6829	0.0191	20,153
Post-COVID	Decision Tree	0.6767	0.0177	20,153
Post-COVID	QDA	0.6759	0.0183	20,153
Post-COVID	Random Forest	0.6716	0.0193	20,153
Post-COVID	SVM (30k subsample)	0.5935	0.0141	20,153
Post-COVID	KNN (30k subsample)	0.5205	0.0101	20,153

Pre-COVID	Logistic Regression (baseline)	0.6753	0.0193	18,561
Pre-COVID	Gradient Boosting	0.6806	0.0204	18,561
Pre-COVID	LDA	0.6733	0.0193	18,561
Pre-COVID	Decision Tree	0.6653	0.0178	18,561
Pre-COVID	QDA	0.6649	0.0184	18,561
Pre-COVID	Random Forest	0.6622	0.0190	18,561
Pre-COVID	SVM (30k subsample)	0.5851	0.0126	18,561
Pre-COVID	KNN (30k subsample)	0.5352	0.0108	18,561

6.4 Robustness

A potential concern is that excluding transitions from employment to “not in labor force” (NILF) could affect the estimated EU risk structure, since NILF may include both voluntary exits and discouraged workers. As a robustness check, we expand the sample to include E to NILF transitions and verify whether our core findings on EU transitions are sensitive to this choice.

Including NILF increases the sample size modestly and yields an E to NILF rate of about 2.60%. Importantly, the EU transition rates by period remain very similar to the baseline estimates, and the relative rankings of key predictors like education and part-time risk remains stable. This supports the approach of focusing on the employed-to-unemployed transition as the primary definition of job loss risk.

Table 3: EU Rate — Baseline vs NILF-Included

Period	EU Rate (baseline)	EU Rate (incl. NILF)	Diff (pp)	N (baseline)	N (with NILF)
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Pre-COVID	0.0092	0.0090	-0.024	928079	952636
COVID	0.0265	0.0258	-0.075	275613	283623
Post-COVID	0.0093	0.0090	-0.024	1007682	1034214

7. Conclusion

This study investigates near-term job-loss risk using linked monthly CPS microdata from 2018 to 2023. We define job loss as a one-month transition from employment at month t to unemployment at month $t+1$, and examine how both the level and structure of job-loss risk differ across pre-COVID, COVID, and post-COVID periods. Our first key finding is that job-loss risk rises sharply during COVID. The monthly EU transition rate increases from approximately 0.9% in pre-COVID to 2.7% during COVID, then returning close to its pre-pandemic level in the post-COVID period. This pattern is visible both in aggregate period averages and in the monthly time series. The magnitude of this increase suggests how quickly a macroeconomic shock can translate into elevated unemployment transitions even at a one-month time.

Second, we document strong and persistent heterogeneity in job-loss vulnerability. Education is consistently protective that higher educational level substantially reduces job-loss risk across all periods. Younger workers and part-time workers face higher EU transition rates, indicating that weaker labor market attachment and lower human-capital buffers correlate with higher short-term unemployment transitions. These descriptive and regression-based results are consistent with standard labor market intuition and provide a coherent baseline profile of vulnerability.

Third, most importantly, we find evidence that the structure of job-loss risk changes across periods. While some predictors remain stable in sign, other relationships shift during the COVID period. For example, the association between gender and job-loss risk differs during COVID relative to pre- and post-COVID, suggesting that period-specific vulnerabilities may reflect pandemic-era sectoral shocks and household constraints. Differences across worker classes also become more obvious during COVID, reinforcing the conclusion that macroeconomic conditions can reconstruct risk profiles across categories of the workforce. From a prediction perspective, our models achieve moderate discrimination but face limitations inherent to rare-event classification. Because EU transitions occur roughly 1% of the time, performance metrics such as PR-AUC remain low in absolute magnitude even when the model captures meaningful signal. Using time-based evaluation provides a more realistic measure of predictive usefulness, and calibration helps interpret predicted risk probabilities as operational risk levels rather than deterministic forecasts.

Finally, robustness checks indicate that our primary conclusions are not driven by a narrow definition of job loss. When we include transitions from employment to NILF, the overall EU transition rates and the ranking of key risk factors remain largely stable, supporting the baseline focus on employed-to-unemployed transitions as the main outcome.

Several limitations may motivate future work. First, CPS month-to-month linking is imperfect, unmatched cases may cause selection. Second, our predictor set is intentionally parsimonious and does not fully exploit higher-dimensional information such as detailed occupation and industry interactions, local macro conditions, or longer-term transitions. Extending the feature set and integrating local labor market indicators could improve both explanatory power and predictive accuracy. Third, our results are predictive and associational rather than causal, so the estimated relationships should not be interpreted as policy effects without additional

identification strategies. In conclusion, the findings show that the COVID shock substantially increased near-term job-loss risk and altered which characteristics best predict transitions into unemployment. The persistent education and part-time gradients show stable vulnerability, and period-dependent shifts emphasize the value of comparing predictive structure across different economic conditions.

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